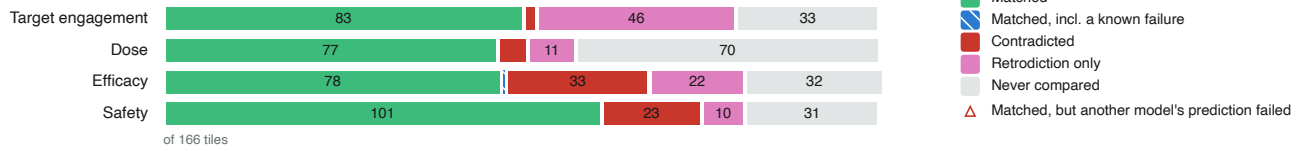
a



Panel b: the top 15 diseases by global DALYs in 2021 (WHO Global Health Estimates), each with the mechanism that had the most programs.

b

		Target engagement	Dose	Efficacy	Safety
#1	**acute coronary syndrome** APOA1 replacement / gene therapy (3)	**2015** ApoE-/- mice infused with apoA-I Milano/phospholipid	**2017** CER-001 4-week repeat-dose animal toxicology (every second day)	**2024** Common APOA1 variants raising apoA-I (Copenhagen City… △ MDCO-216 no regression vs…	**2016** Nonclinical toxicology of CSL112 Model data unpublished
#2	**type 2 diabetes mellitus** GLP1R agonists (12)	**2011** Cynomolgus monkeys and r… Glucose-dependent insulin rose in humans	**2011** Rat and cynomolgus PK (dulaglutide)	**2022** GLP1R Gly168Ser genotype, GLP-1 infusion clamp in healthy volunteers	**2014** 1-year cynomolgus thyroid C-cell study (dulaglutide)
#3	**COPD** CHRM3 antagonists (6)	**2018** Human isolated bronchi:… Tissue offset >10 h, patients bronchodilated 24 h	**2016** Pfizer guinea-pig tracheal-strip relaxation (slow M3 off-rate) and in vivo…	**2018** Human isolated bronchi:… Trough FEV1 primary met, as predicted	**2016** Rat bronchoprotection and antisialagogue; dog bronchospasm △ Stopped; preclinical unexplained
#4	**HIV** HIV RT (NNRTI) inhibitors (3)	**2015** HIV-1 culture vs wild-type a… Doravirine monotherapy cut HIV RNA 1.3 log10	**2020** In-silico doravirine PK model with rifabutin from the single-dose DDI, setting 100 mg twice daily	**2017** HIV-1 culture vs wild-type and NNRTI-resistant virus, resistance… Phase 3 non-inferior to darunavir/r	**2011** Rodent and monkey tox as filed for fosdevirine and doravirine Seizures no tox species predicted
#5	**NSCLC** EGFR inhibitors (18)	**2017** T790M/sensitising-mutant NSCLC cells and mice	**2019** PC9 brain-metastasis mice; monkey osimertinib PET	**2023** Patient-derived lung cancer organoids △ EXCLAIM-2 fails primary PFS	**2021** hERG, rat and dog tox, dog telemetry QT prolongation the package cleared
#6	**migraine disorder** CGRP receptor antagonists (4)	**2020** Rhesus capsaicin-induced dermal vasodilation (CIDV) Rhesus EC50 3.2 vs human 2.6 nM	**2014** Rhesus CIDV PK/PD target Predicted 20 mg; needed 200 mg	**2020** Rat sumatriptan-primed… Rat allodynia reproduces ubrogepant efficacy	**2026** Mouse MCAO stroke, gepant pretreatment △ Animal tox missed MK-3207 liver
#7	**Alzheimer disease** BACE1 inhibitors (4)	**2015** PDAPP mice and cannulated beagle dogs (brain/CSF Aβ) Human CSF Aβ fell as in dogs	**2015** Dog CSF Aβ PK/PD scaled to humans No checkable prediction	**2021** PDAPP (APP V717F) transgenic mice No change in tau PET	**2018** Mice chronically dosed with BACE1 inhibitors △ Liver injury missed by tox
#8	**asthma** PTGDR2 antagonists (9)	**2016** Human eosinophils and Th2 cells in vitro	**2016** Human eosinophils and Th2 cells in vitro	**2011** Allergen challenge	**2012** Off-target selectivity panels; animal toxicology unpublished Model data unpublished
#9	**breast cancer** CDK4/6 inhibitors (6)	**2013** Human tumour xenografts in mice (phospho-Rb Ser780)	**2017** Rodent brain penetration and intracranial xenograft (abemaciclib)	**2020** Human breast cancer cell-li… PALLAS adjuvant palbociclib fails iDFS	**2017** Dog cardiovascular telemetry (QTc) with hERG assays △ Missed liver injury and clots
#10	**osteoarthritis** MAPK14 inhibitors (3)	**2012** Rat iodoacetate joint degeneration + carrageenan thermal hyperalgesia (SB-203580, VX-745)	**2009** Rat and cyno endotoxin TNF-alpha PD Patient result unclear	**2012** Rodent rheumatoid-arthritis models, ex-vivo LPS whole-blood cytokine PD (ARRY-797)	**2014** Rat repeat-dose intra-articular toxicology (FX005) Not measured in patients
#11	**colorectal cancer** KDR (VEGFR) inhibitors (3)	**2014** Human tumour cell-line xenografts in nude rodents (incl. VEGF-challenge pVEGFR2 readout)	**2014** Human tumour cell-line xenografts in nude rodents (incl.… △ Start dose toxic; animal margin…	**2017** Human colorectal patient-derived xenografts (PDX) i… △ BATON-CRC PFS futility vs…	**2023** Rat, dog and monkey… Dogs silent; 14% grade 3-4 hypertension
#12	**chronic hepatitis B virus infection** HBV RNA antisense / siRNA (3)	**2020** HBV-genome mice (pHBV transient/transgenic, later AAV-HBV)	**2018** HBV-infected human liver-chimeric mice No checkable prediction	**2024** HBV-genome mice (pHBV transient/transgenic, later AAV-HBV)	**2026** HBV-genome mice (pHBV transient/transgenic, later AAV-HBV) No ALT rise in mice, as in volunteers
#13	**hepatocellular carcinoma** VEGFR multikinase inhibitors (4)	**2015** Human HCC cell-line and PDX xenografts Not measured in patients	**2015** Human HCC cell-line and PDX xenografts	**2020** Mouse HCC with diverse transposon-delivered genetic drivers, lenvatinib vs sorafenib susceptibility	**2018** GLP repeat-dose toxicology in rat plus dog or monkey
#14	**schizophrenia** PDE10A inhibitors (5)	**2019** Rat and monkey ketamine-challenge phMRI/EEG reversal	**2013** Rodent occupancy dosing	**2013** Rodent CAR / psychostimulant hyperlocomotion Monotherapy PANSS not met	**2017** Rat catalepsy, prolactin and glucose vs D2 antagonists EPS in patients at clinical dose
#15	**gastric cancer** PARP1/2 inhibitors (3)	**2023** Gastroesophageal PDX, BRCA2-mutant, PARP inhibitors Not measured in patients	**2023** Gastroesophageal PDX, BRCA2-mutant, PARP inhibitors No checkable prediction	**2026** Gastroesophageal PDX, BRCA2-mutant, PARP inhibitors △ Pamiparib maintenance missed…	**2017** Rat and dog general toxicology, up to 26 weeks (olaparib)